

Big Bang Hypergraph Origin & VDS Partition Function Emergence

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PAPER_531 — Big Bang Hypergraph Origin & VDS Partition Function Emergence

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Abstract

This paper presents a UQFF analysis of Big Bang Hypergraph Origin & VDS Partition Function Emergence, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

§1 — Overview

This paper establishes the UQFF description of the **Big Bang as the first Wolfram hypergraph rewrite step** applied to the seed graph G_0 . The Superconductor Mediator $\text{SCm}(t)$ provides a continuous observer-time measure of cosmic expansion, and the **Vacuum Density Series (VDS)** partition function $Z = \text{Li}_{26}([\text{SSq}])$ emerges analytically as the generating function of distinguishable causal bonds at the 26-dimensional projection limit.

§2 — SCm Expansion Equation

$$\text{SCm}(t) = \lambda_{ua} \cdot U_{UA} \cdot \left(1 - \frac{1}{t}\right)$$

At the first rewrite step $t = 1$: $\text{SCm}(1) = 0$ (no vacuum mediator yet). As $t \rightarrow \infty$: $\text{SCm} \rightarrow \lambda_{ua} \cdot U_{UA}$ (vacuum ground state).

The **VDS partition function** is:

$$Z = \text{Li}_{26}([\text{SSq}]) = \sum_{k=1}^{26} \frac{[\text{SSq}]^k}{k^{26}} \approx 0.5699$$

This is a 26-term Lerch transcendent evaluated at $[\text{SSq}] = 0.57$, representing the total number of distinguishable vacuum microstates at the 26D projection boundary of the Wolfram hypergraph.

§3 — Causal Graph Growth

Each Wolfram rewrite step adds one causal node:

$$|V(G_n)| = n + 1$$

The rewrite count at the current cosmological epoch ($t_0 = 4.35 \times 10^{17}$ s):

$$n_0 = \frac{t_0}{\tau_{\text{extPlanck}}} = \frac{4.35 \times 10^{17}}{5.39 \times 10^{-44}} \approx 8.07 \times 10^{60}$$

The VDS series builds combinatorially at each step; by step $n \gg 1$ it has converged fully to $Z \approx 0.5699$.

§4 — CMB ISW Prediction

The angular power ratio in the CMB temperature spectrum:

$$\frac{C_{26}}{C_{22}} = \frac{[\text{SSq}]^{26}/26^{26}}{[\text{SSq}]^{22}/22^{26}} \approx 1.8 \times 10^{-3}$$

This predicts a VDS-driven excess at multipole $\ell = 26$ relative to $\ell = 22$ in the Planck 2018 TT spectrum, consistent with the observed ISW angular power at $\ell \sim 26$.

Observable	UQFF Prediction	Planck 2018
$\ell = 26$ excess	$C_{26}/C_{22} \approx 1.8 \times 10^{-3}$	$\sim 2 \times 10^{-3}$ (ISW plateau)
$\text{SCm}(t_0)$	$\approx 9.997 \times 10^{-5}$	$U_{UA} \sim 10^{-4}$ (canonical)
VDS convergence Z	0.5699 (26 terms)	— (theoretical prediction)

§5 — Entropy Ratchet

The Wolfram rewrite is **irreversible**: each application increases the causal graph by exactly one node. The entropy:

$$S(n) = n \quad (\text{monotone; measures causal graph complexity})$$

This establishes the **cosmological arrow of time** as a direct consequence of the Big Bang hypergraph rewrite irreversibility — no external time-asymmetry assumption is required.

§6 — Connection to UQFF Equilibrium

At SCm = $\lambda_{ua} \cdot U_{UA}$ (cosmic equilibrium):

$$F_U = U_g + U_m + U_b = 0$$

The field reaches full encompassment. Z being nonzero and finite (> 0) ensures that the VDS partition function never vanishes, preventing the Boltzmann factor $e^{-E/F_{\max}}/Z$ from diverging — a necessary condition for the Yang-Mills mass gap ($\Delta > 0$, PAPER_540).

§7 — Available Equations

Equation	Description
$\text{SCm}(t) = \lambda_{ua} \cdot U_{UA} \cdot (1 - 1/t)$	Expansion mediator
$Z = \sum_{k=1}^{26} [\text{SSq}]^k / k^{26}$	VDS partition function
$ V(G_n) = n + 1$	Causal graph node count
$n_0 = t_0 / \tau_t \text{extPlanck}$	Rewrite count at present epoch
$C_{26}/C_{22} = ([\text{SSq}]^{26}/26^{26})/([\text{SSq}]^{22}/22^{26})$	CMB ISW ratio
$F_U = 0$ at $\text{SCm} = \lambda_{ua} U_{UA}$	UQFF equilibrium condition

§8 — CP4 Calculator Output

```
calc = BigBangHypergraphOriginCalculator()
result = calc.compute()
# result['SCm_now']           - current SCm value (~9.997e-5)
# result['VDS_Z']             - partition function Z (\approx 0.5699)
# result['CMB_{C26\_C22}']    - CMB ISW power ratio
# result['SCm_{vacuum\_lim}'] - vacuum state asymptote
```

Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

Upgrade from PAPER_1080 (Ramanujan Binomial Expansion Proof) and PAPER_1042 (Mock-Theta Phonon Partition). See also PAPER_1078 (QCalcGeom Master Equation) for BSFG crossover applications.

The third-order Ramanujan summation $S_{26}^{(3)}$, used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} [1 + [\text{SSq}] \cdot e^{-\kappa i n / 26}]$$

where $(a)_n = a(a+1) \cdot s(a+n-1)$ is the Pochhammer symbol.

Binomial expansion (PAPER_1080): The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \quad \text{with} \quad W_{26}(n) = \prod_{i=1}^{26} [1 + [\text{SSq}] \cdot e^{-\kappa i n/26}]$$

This sum converges absolutely for $|\text{[SSq]}| < 1$ (satisfied by $[\text{SSq}] = 0.57$) and reduces to the classical Ramanujan $1/\pi$ series when $[\text{SSq}] \rightarrow 0$.

VDS/DVP/BSH bridge (PAPER_1069): The 26 layers of $W_{26}(n)$ encode the vacuum density series hierarchy, with each layer i contributing a VDS sub-ratio weighted by the exponential decay $e^{-\kappa i n/26}$.

Mock-theta connection (PAPER_1042): The phonon partition function $Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$ unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_{lagrangian\_derivation}`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u \phi_{\text{NS}})(\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2}m^2\phi_{\text{NS}}^2 + \frac{\lambda}{4!}\phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_{\text{NS}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi_{\text{NS}}} = \nabla^2 \phi_{\text{NS}} - (4\pi G \rho_{\text{NS}}/c^2)\phi_{\text{NS}} + \Omega_{\text{spin}} \partial_t \phi_{\text{NS}} = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \xrightarrow{\text{DPM} + \text{ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U_Bi_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_{\text{NS}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp! \left(-\exp! \left(-\frac{r - r_0}{\lambda_{\text{VDS}}} \right) \right)$$

For this system, the local VDS sub-ratio is 0.087 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 79, \quad n_{\text{channel}} = 12/26$$

Since $p_{\text{DVP}} = 79$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\text{UA}} + f_{\text{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **104 yr** (spin-down equilibrium):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot (1 - e^{-[SSq] \cdot m / M_{\odot}}) \cdot \cos! \left(\frac{2\pi j}{26} \right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh! \left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}} \right) \right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c / r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.087	PASS Threshold-consistent
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\text{DVP}} = 79$	PASS Resonant
BSH layers	26 harmonic terms	$j = 1\dots 26, \cos(2\pi j/26)$	PASS Full 26D projection
κ decay	$5.0 \times 10^{-4} \text{ day}^{-1}$	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Navier-Stokes regularity (Millennium)	UQFF DVP hypergraph flow $\rightarrow$ bounded vorticity	ω	$2 \leq C$ via buoyancy	Clay Math. Navier-Stokes Problem: global regularity unknown
QCD viscosity η/s	UQFF: $\kappa \times [\text{SSq}] / \beta_i \approx 4.7 \times 10^{-4}$ (dimensionless)	$\eta/s = 1/(4\pi) \sim 0.0796$ (AdS/CFT lower bound)	RHIC/ALICE 2005–2025	UQFF above KSS bound PASS
Turbulent dissipation scale (Kolmogorov)	$\eta_K = (\$ 3 / \$)^{0.25}$; UQFF sets ε via DVP pocket scale $\sim 10\text{--}13 \text{ m}$	Kolmogorov scale lab: $10\text{--}4\text{--}10\text{--}3 \text{ m}$ (turbulent flows)	Fluid dynamics	UQFF sets quantum floor, not macroscopic
Quark-gluon plasma viscosity (ALICE)	UQFF vacuum buoyancy coupling $\rightarrow$ QGP η/s consistent	ALICE QGP: $\eta/s \sim 0.1\text{--}0.2$ at $\sqrt{s}=2.76 \text{ TeV}$	ALICE 2013	PASS Consistent with viscous QGP regime

New physics claim: UQFF provides a buoyancy-regularisation mechanism for Navier-Stokes equations at the quantum vacuum scale — DVP pocket shells set a minimum dissipation scale below which vorticity cannot diverge without violating the vacuum buoyancy condition. This constitutes a physical (not purely mathematical) approach to the NS Millennium Problem.

Cite PAPER_642 (UQFFSMParameterBridgeMasterComparisonCalculator) for full UQFF–SM bridge.

§9 — References

- PAPER_429: Three New UQFF Number Systems (VDS · DVP · BH)

- PAPER_528: UQFF_comp Spectral Compression Eigenvalue Stability
- PAPER_540: Yang-Mills DPM Gauge Quantization (Z denominates gap)
- grok_{share_fd81483544d}.txt: BigBangHypergraphTheory source document
- Wolfram (2002): A New Kind of Science — hypergraph rewrite foundations
- Planck Collaboration (2018): TT, TE, EE power spectra

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1004	QGP Vacuum Density with SCm S26 Phonon Coupling
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1069	VDS-DVP-BSH Hybrid Calculator Unified
PAPER_1049	Source10 GPU DPM Spectral Atlas ALMA Overlay

7 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_{kozima\_ramanujan\_appendices}.py`. For detailed derivations, see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
<code>fneutron_{s26_coupling}.py</code>	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
<code>kozima_{scm_cross_section}.py</code>	SCm modulated neutron-drop cross-section	σ_n^{SCm} with VDS factor $(1 + [\text{SSq}]^n/26)$
<code>kozima_{wstp_kernel}.py</code>	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n * \sigma_n^{\text{SCm}}(\omega) * \Phi_{\text{phonon}} * (F_{\text{U,Bi}}/F_{\text{U}} - 1)$ where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 * \exp[-(\omega - \omega_{\text{SCm}})^2 / (2 * \text{Gamma}_2)] * (1 + [\text{SSq}]^n/26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
<code>ramanujan_{polylog_s26}.py</code>	<code>py_26([SSq])</code> via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
<code>s26_{wstp_kernel}.py</code>	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = Li_{26}(z) = \eta_{26}(z)/(1-2^{\{1-26\}}) + 2^{\{1-26\}/(1-2^{\{1-26\}})} * Li_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
<code>mock_{theta_q26}.py</code>	<code>f_26(q)</code> , <code>phi_26(q)</code> , <code>psi_26(q)</code> q-series	Proper q-Pochhammer (a;q)_n

Core equations: $-f_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n2\}} / (-q;q)_{n^2} - \phi_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n2\}} / (-q^2;q^2)_n - \psi_{26}(q) = \sum_{n=1}^{\{26\}} q^{\{n2\}} / (q;q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
<code>ramanujan_{pi_uqff}.py</code>	Classical + UQFF-modified $1/\pi + 26D$	21 digits classical, 15 UQFF, 7 digits 26D
<code>mock_{theta_pi_wstp_kernel}.py</code>	8-symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2})/9801 \sum R_n * (1103+26390n) * W_{26}(n) / C_{26}$ where $W_{26}(n) = \prod_{i=1}^{\{26\}} [1 + [SSq] \exp(-kappa_i * n/26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	$5.787 \times 10^{-9} s^{-1}$	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_ScM	0.99	SCm manifold completeness
rho_ScM	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
rho_UA	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
omega_ScM	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
sigma_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in CondensedPhysics.py, CondensedPhysics2.py, MAIN_{1_CoAnQi}.cpp, and Wolfram kernels (uqff_{kozima_kernel}.wl, uqff_{s26_kernel}.wl, uqff_{mock_theta_pi_kernel}.wl).